

OPERATING SYSTEMS

III B. TECH- I SEMESTER

Course Code	Category	Hours / Week			Credits	Maximum Marks		
A5CS13	PCC	L	T	P	C	CIE	SEE	Total
		3	1	-	4	30	70	100

COURSE OBJECTIVES

1.

To explain main components of OS and their working

2.

To familiarize the operations performed by OS as a resource Manager

3.

To familiarize with Dead Locks

4.

To impart various scheduling policies of OS

5.

To teach the different memory management techniques.

COURSE OUTCOMES

1.

Analyze the different structures and services of operating system.

2.

Analyze various algorithms used for OS services with respect to defined/chosen criteria.

3.

Solve the resource allocation and sharing problems.

4.

Assess different methods to solve OS problems.

5.

Analyze the memory management approaches of operating systems.

UNIT-I	OPERATING SYSTEMS OVERVIEW	CLASSES: 14
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OPERATING SYSTEMS OVERVIEW: Introduction, operating system operations, process management, memory management, storage management, protection and security, distributed systems.

OPERATING SYSTEMS STRUCTURES: Operating system services and systems calls, system programs, operating system structure, operating systems generations

UNIT - II	PROCESS MANAGEMENT	CLASSES: 16
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PROCESS MANAGEMENT: Process concepts, process state, process control block, scheduling queues, process scheduling, Scheduling algorithms, multithreaded programming, threads in UNIX, comparison of UNIX and windows.

CONCURRENCY AND SYNCHRONIZATION: Process synchronization, critical section problem, Peterson's solution, synchronization hardware, semaphores, classic problems of synchronization, readers and writers problem, dining philosophers problem, monitors, synchronization examples(Solaris).

UNIT - III	DEAD LOCKS	CLASSES: 14
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DEADLOCKS: System model, deadlock characterization, deadlock prevention, detection and avoidance, recovery from deadlock banker's algorithm.

MEMORY MANAGEMENT: Swapping, contiguous memory allocation, paging, structure of the page table, segmentation, virtual memory, demand paging, page-replacement algorithms, allocation of frames, thrashing.

UNIT - IV	FILE SYSTEM	CLASSES: 12
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FILE SYSTEM: Concept of a file, access methods, directory structure, file system mounting, file sharing, protection.

File system implementation: file system structure, file system implementation, directory implementation, allocation methods, free-space management, efficiency and performance.

UNIT - V	MASS STORAGE STRUCTURE	CLASSES: 14
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MASS STORAGE STRUCTURE: overview of mass storage structure, disk structure, disk attachment, disk scheduling algorithms, swap space management, stable storage implementation, tertiary storage

structure.

I/O: Hardware, application I/O interface, kernel I/O subsystem, transforming I/O requests to hardware operations, streams, performance

TEXT BOOKS

1. Operating System Concepts, Abraham Silberschatz, Peter Baer Galvin, Greg Gagne (2006).
2. Operating System Principles, Abraham Silberschatz, Peter Baer Galvin, Greg Gagne (2006), 7th edition, Wiley India Private Limited, New Delhi.

REFERENCE BOOKS

1. Operating Systems, Internals and Design Principles, Stallings (2006), 5th edition, Pearson Education, India.
2. Modern Operating Systems, 2nd edition, Andrew S. Tanenbaum (2007), Prentice Hall of India, India
3. Operating systems, Deitel & Deitel (2008), 3rd edition, Pearson Education, India.

WEB LINKS

1. <https://nptel.ac.in/courses/106/102/106102132/>
2. https://onlinecourses.nptel.ac.in/noc21_cs88/preview